Formula sheet first partial IS exam P2 2023

This sheet only contains formulas, no instructions on how to use them.

Propositional Logic

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{de Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{de Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Modens Ponens $\dfrac{\alpha \Rightarrow \beta, \alpha}{\beta}$

And-elimination: from a conjunction any conjunct can be inferred: $\dfrac{\alpha \wedge \beta}{\alpha}$

Logical equivalences can be used as rules:

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

RDFS Reasoning (you can ignore rules 1-4)

```
5.  s   rdfs:subClassOf   y
    y   rdfs:subClassOf   z  ⟹  s   rdfs:subClassOf   z

6.  p  rdfs:subPropertyOf  q
    q  rdfs:subPropertyOf  r  ⟹  p  rdfs:subPropertyOf  r

7.  s          p           o
    p  rdfs:subPropertyOf  q  ⟹  s          q          o

8.  s          p           o
    p     rdfs:domain      x  ⟹  s       rdf:type      x

9.  s          p           o
    p      rdfs:range      x  ⟹  o       rdf:type      x

10. s      rdf:type        o
    o      rdfs:subClassOf t  ==>  s      rdf:type      t
```